Fall 2026

Bayesian Techniques in Macroeconomics

Université Paris Dauphine – PSL
Master in Quantitative Economics, second year

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Chapter 2

A Production Economy

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Handout Objectives

- Solving a rational-expectations model.
- Linearizing a system of equations.
- Extending a model and checking its solution.

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Session roadmap

Economic intuition. Capital links current investment to future production. **Bayesian/econometric intuition.** The equilibrium solution becomes the transition equation used by the Kalman filter. **Model.** Derive the steady state, log-linearize, and verify determinacy. **MATLAB/Dynare exercise.** Compare analytical and Dynare solutions, then interpret moments and IRFs. **Takeaway.** Never estimate a DSGE model before residual and Blanchard–Kahn checks pass.

1. A production economy

Our economy is characterized by a representative household that consumes, produces goods and accumulates physical capital used in production. There is no variable labour input in this economy: the household's labour supply is inelastic to macroeconomic fluctuations while technology does not involve hours worked. Households are infinitely lived, and the economy is populated by a continuum of households $j \in [0, 1]$ with identical initial capital. Assume $0 < \beta < 1$, positive consumption, and a transversality condition excluding overaccumulation of capital.

The household faces the following maximization problem:

$$\max_{C_{jt}, K_{jt}} \mathbb{E}_t \sum_{\tau=0}^{+\infty} \beta^\tau \log\left(C_{jt+\tau}\right)$$
$$s.t. : C_{jt} + K_{jt} = A_t K_{jt-1}^{\alpha} + (1-\delta) K_{jt-1}$$

The budget constraint combines the resource constraint with the accumulation of physical capital. First, GDP is allocated as follows:

$$C_{jt} + I_{jt} = Y_{jt}. \tag{1}$$

The total income Y_{jt} is spent on consumption goods C_{jt} and investment goods I_{jt}.

The law of motion of capital reads as in the Solow [1956]'s model:

$$I_{jt} = K_{jt} - (1-\delta) K_{jt-1} \tag{2}$$

Investment is determined by the gap between current capital K_{jt} and depreciated capital $(1-\delta) K_{jt-1}$. In the latter expression, the parameter $0 \leq \delta \leq 1$ denotes the depreciation rate of capital.

The standard Cobb-Douglas technology is determined by:

$$Y_{jt} = A_t K_{jt-1}^{\alpha} \tag{3}$$

where $0 < \alpha < 1$ is the share of capital involved in the production process of new goods Y_{jt} while A_t is a productivity shock which temporarily affects the society's ability to produce goods, it affects all the households in our economy in the same way. The production combines technology and capital K_{jt-1} bought in $t-1$ because investment chosen at $t-1$ becomes productive at t. This is the standard one-period installation convention, not a multi-stage time-to-build mechanism. Concerning the productivity shock, it is assumed to follow a simple AR(1) process given by:

$$\log\left(A_t\right) = \rho_A \log\left(A_{t-1}\right) + \eta_t^A, \tag{4}$$

where $0 \leq \rho_A < 1$ is the autocorrelation coefficient of the shock and η_t^A is an innovation distributed normally with zero mean and finite variance σ_A^2 such that $\eta_t^A \sim \mathcal{N}(0, \sigma_A^2)$.

Exercise 1: A production economy

Consider the production economy model.

1. Set the Lagrangian problem.
2. Find the first-order conditions. Get rid of the Lagrangian multiplier to obtain the Euler equation on physical capital.
3. Write the three equations of the model. Apply aggregation to find the general equilibrium conditions such that $C_t = \int_0^1 C_{jt}\mathrm{d}j, K_t = \int_0^1 K_{jt}\mathrm{d}j$ and $Y_t = \int_0^1 Y_{jt}\mathrm{d}j$. Find the steady state.
4. Linearize the three equations.

1.1 Equilibrium and the local benchmark

A unit of capital costs one unit of current consumption. Next period it returns its marginal product and undepreciated capital. Optimal choices imply

$$\lambda_{jt} = C_{jt}^{-1}, \qquad \lambda_{jt} = \beta E_t[\lambda_{j,t+1}(\alpha A_{t+1} K_{jt}^{\alpha-1} + 1 - \delta)].$$

The return and future marginal utility are jointly random, so their product must remain inside the expectation. With identical initial capital and a common shock, the symmetric equilibrium satisfies

$$\begin{aligned} C_t + K_t &= A_t K_{t-1}^{\alpha} + (1-\delta)K_{t-1}, \\ C_t^{-1} &= \beta E_t[(\alpha A_{t+1} K_t^{\alpha-1} + 1 - \delta)/C_{t+1}], \\ \log A_t &= \rho_A \log A_{t-1} + \eta_t^A. \end{aligned}$$

Without symmetry, integrating K_{jt}^{α} would not generally yield K_t^{α}. In the deterministic steady state, $\bar{A} = 1$ and

$$\bar{K} = \left(\frac{\beta^{-1} - 1 + \delta}{\alpha}\right)^{1/(\alpha-1)}, \quad \bar{Y} = \bar{K}^{\alpha}, \quad \bar{I} = \delta\bar{K}, \quad \bar{C} = \bar{Y} - \bar{I}.$$

Define $\hat{c}_t = \log(C_t/\bar{C})$, $\hat{k}_t = \log(K_t/\bar{K})$ and $\hat{a}_t = \log A_t$. Hats are log deviations in decimal units. First-order expansion of feasibility gives

$$\bar{C}\hat{c}_t + \bar{K}\hat{k}_t = \bar{K}^{\alpha}(\hat{a}_t + \alpha\hat{k}_{t-1}) + (1-\delta)\bar{K}\hat{k}_{t-1}.$$

Writing $R_{t+1} = \alpha A_{t+1} K_t^{\alpha-1} + 1 - \delta$ and using $\beta\bar{R} = 1$ gives the Euler expansion

$$-\hat{c}_t = E_t[-\hat{c}_{t+1} + \beta\alpha\bar{K}^{\alpha-1}(\hat{a}_{t+1} + (\alpha-1)\hat{k}_t)].$$

Exercise 1 asks you to reconstruct the optimization and aggregation, not merely copy these formulas. These steps explain the coefficients used below.

2. Solving a linear rational-expectations model

2.1 A general solution

Recall that in a general form, a model is a set of nonlinear equations:

$$\mathbb{E}_t \left[f_\theta \left(x_{t+1}, x_t, x_{t-1}, \epsilon_t \right) \right] = 0, \tag{5}$$

where θ is the set of structural parameters, x_t are endogenous variables, ϵ_t are Gaussian innovations and $f_\theta(\bullet)$ is the set of equations.

As discussed in the previous handout, we apply linearization methods to $f_\theta()$ to obtain the set of linear equations:

$$\mathbb{E}_t \left[\hat{f}_\theta \left(\hat{x}_{t+1}, \hat{x}_t, \hat{x}_{t-1}, \epsilon_t \right) \right] = 0 \tag{6}$$

A classification of variables and equations. In order to avoid singular matrices, we must distinguish between three types of variables: (i) state variables $\hat{s}_t$, (ii) jump variables $\hat{\jmath}_t$ and (iii) exogenous shock processes $\hat{z}_t$.[2] Predetermined states carry information inherited from past choices. Jump variables can adjust when new information arrives, although they may inherit persistence from the states. Exogenous shock processes evolve according to their own specified laws. Note that Dynare automatically detects states and jump variables and reports their respective numbers in the command window of MATLAB.

Given this classification across endogenous variables $\hat{x}_t = [\hat{s}_t, \hat{z}_t, \hat{\jmath}_t]'$, we also must distinguish between three types of equations in our linearized model $\hat{f}_\theta()$:

$$0 = E_t \left[\begin{array}{cccc} A_0 \hat{s}_t + & A_{-1} \hat{s}_{t-1} + & B_0 \hat{\jmath}_t + & C_0 \hat{z}_t \end{array} \right], \tag{7}$$

$$0 = E_t \left[\begin{array}{cccccc} D_1 \hat{s}_{t+1} + & D_0 \hat{s}_t + & D_{-1} \hat{s}_{t-1} + & M_1 \hat{\jmath}_{t+1} + & M_0 \hat{\jmath}_t + & L_1 \hat{z}_{t+1} + \quad L_0 \hat{z}_t \end{array} \right], \tag{8}$$

$$\hat{z}_t = Z \hat{z}_{t-1} + \epsilon_t, \; \epsilon_t \sim \mathcal{N}(0, Q). \tag{9}$$

The first set of equations stacks equations with no expectation terms, the second set of equations must feature at least one forward looking variable, and the last set of equation are exogenous shock processes.

The functional form of the solution. We are looking for a recursive solution of our model:

$$\hat{x}_t = F \hat{x}_{t-1} + G \hat{z}_t, \tag{10}$$

where F and G are yet two unknown matrices. Note that this solution must hold for transitions from $t-1$ to t and from t to $t+1$ consistently with our linearized system $\hat{f}_\theta()$. Using our classification, this solution can be expressed as two separate processes:

$$\left[\begin{array}{c} \hat{s}_t \\ \hat{\jmath}_t \end{array} \right] = \left[\begin{array}{cc} F_s & 0_{n_j} \\ F_j & 0_{n_j} \end{array} \right] \left[\begin{array}{c} \hat{s}_{t-1} \\ \hat{\jmath}_{t-1} \end{array} \right] + \left[\begin{array}{c} G_s \\ G_j \end{array} \right] \hat{z}_t \tag{11}$$

The unknowns in the previous system are the matrices F_s, F_j, G_s and G_j. Another way to present the solution reads as follows:

$$\hat{s}_t = F_s \hat{s}_{t-1} + G_s \hat{z}_t, \tag{12}$$

$$\hat{\jmath}_t = F_j \hat{s}_{t-1} + G_j \hat{z}_t. \tag{13}$$

[2] Note that Dynare has a different classification for endogenous variables as exogenous variables are also treated as endogenous state processes.

Solution of the system. To obtain the unknowns of our transition equation, consider our first set of equations in Equation 7. We must use equations (12-13) to replace $\hat{s}_t$ and $\hat{\jmath}_t$ as follows:

$$\begin{aligned}
0 = &\ A_0\hat{s}_t + \ A_{-1}\hat{s}_{t-1} + \ B_0\hat{\jmath}_t + \ C_0\hat{z}_t \\
0 = &\ A_0\left(F_s\hat{s}_{t-1} + G_s\hat{z}_t\right) + \ A_{-1}\hat{s}_{t-1} + \ B_0\left(F_j\hat{s}_{t-1} + G_j\hat{z}_t\right) + \ C_0\hat{z}_t \\
0 = &\ \underbrace{\left[A_0F_s + A_{-1} + B_0F_j\right]}_{\text{condition (1)}}\hat{s}_{t-1} + \underbrace{\left[A_0G_s + B_0G_j + C_0\right]}_{\text{condition (2)}}\hat{z}_t
\end{aligned}$$

We apply the same substitutions to the second set of equations in Equation 8 to obtain:

$$\begin{aligned}
0 = &\underbrace{\left[D_1\left(F_s\right)^2 + D_0F_s + M_1F_jF_s + D_{-1} + M_0F_j\right]}_{\text{condition (3)}}\hat{s}_{t-1} \\
&+ \underbrace{\left[D_1F_sG_s + D_0G_s + M_1F_jG_s + M_0G_j + D_1G_sZ + M_1G_jZ + L_1Z + L_0\right]}_{\text{condition (4)}}\hat{z}_t,
\end{aligned}$$

These four matrix restrictions determine candidates for F_s, F_j, G_s and G_j when dimensions and rank conditions permit a solution. Counting matrix equations alone is not an existence or uniqueness proof. However in practice, it's not as obvious because condition (3) is a quadratic matrix equation problem of the form $AX^2 + BX + \ C = 0$. There are two implications from this quadratic problem:

1. Solving for the transition matrix F_s is rather difficult, but fortunately Uhlig et al. [1995] provides a solution in his Theorem 3. In addition, Dynare has also its own routine to solve linear model (based on the Schur decomposition).

2. The matrix quadratic can admit several solutions, not necessarily exactly two. A unique stable solution requires the Blanchard–Kahn eigenvalue count and rank condition. Dynare checks these conditions and selects the stable invariant subspace. A scalar example is developed below.

Let us assume that we have followed carefully theorem (3) from Uhlig et al. [1995], which allowed us to solve the quadratic matrix equation problem and pick the stable root of the system. If F_s is known and B_0 is square and invertible, then condition (1) allows us to find F_j as follows:

$$F_j = -\left(B_0\right)^{-1}\left(A_0F_s + A_{-1}\right) \tag{14}$$

Combining conditions (2) and (4) identifies the two remaining unknown matrices:

$$A_0G_s + B_0G_j = -C_0, \tag{15}$$

$$(D_1F_s + D_0 + M_1F_j)G_s + D_1G_sZ + M_0G_j + M_1G_jZ = -L_1Z - L_0. \tag{16}$$

These are simultaneous linear equations for the entries of G_s and G_j. Their matrix products must retain the displayed order. Before Dynare became widespread in the profession, macroeconomists used to linearize their model equations, find the steady state, and fill matrices in Equation 7-8-9 to get the model's dynamics. Changing a single equation would imply to update all the equations. Dynare really simplified the problem by automating the linearization of equations, filling the matrices, finding the solution, selecting the stable root, plotting IRFs, and computing second-moment statistics. Note also that the number of elements in dense square matrices increases quadratically with the number of variables. Numerical solution costs also rise. Thus, most DSGE models were rather small before the use of Dynare.

Dynare formulation. Note that Dynare considers exogenous shock processes as state variables, replacing $\hat{z}_t$ into the system, we obtain:

$$\begin{bmatrix} \hat{s}_t \\ \hat{\jmath}_t \end{bmatrix} = \begin{bmatrix} F_s & 0 \\ F_j & 0 \end{bmatrix} \begin{bmatrix} \hat{s}_{t-1} \\ \hat{\jmath}_{t-1} \end{bmatrix} + \begin{bmatrix} G_s \\ G_j \end{bmatrix} Z\hat{z}_{t-1} + \begin{bmatrix} G_s \\ G_j \end{bmatrix} \epsilon_t, \tag{17}$$

which is formulated in Dynare as:

$$\begin{bmatrix} \hat{s}_t \\ \hat{z}_t \\ \hat{\jmath}_t \end{bmatrix} = \begin{bmatrix} F_s & G_s Z & 0_{n_j} \\ 0_{n_s} & Z & 0_{n_j} \\ F_j & G_j Z & 0_{n_j} \end{bmatrix} \begin{bmatrix} \hat{s}_{t-1} \\ \hat{z}_{t-1} \\ \hat{\jmath}_{t-1} \end{bmatrix} + \begin{bmatrix} G_s \\ I_{n_z} \\ G_j \end{bmatrix} \epsilon_t \tag{18}$$

Or equivalently to the initial form of the policy rule:

$$\hat{x}_t = F\hat{x}_{t-1} + G\epsilon_t, \tag{19}$$

2.2 An application to the production economy

Let's go back to a production economy linearized as follows:

$$\begin{aligned} -\hat{k}_t + k_1\hat{a}_t + k_2\hat{k}_{t-1} - k_3\hat{c}_t &= 0 \\ -\hat{c}_t + E_t\hat{c}_{t+1} - c_1E_t\hat{a}_{t+1} - c_2\hat{k}_t &= 0 \\ -\hat{a}_t + \rho_A\hat{a}_{t-1} + \eta_t^A &= 0 \end{aligned}$$

where $k_1 = \bar{K}^{\alpha-1}$, $k_2 = \bar{K}^{\alpha-1}\alpha + (1-\delta)$, $k_3 = \bar{C}/\bar{K}$, $c_1 = \beta\alpha\bar{K}^{\alpha-1}$, $c_2 = \beta\alpha\bar{K}^{\alpha-1}(\alpha-1)$ and $E_t\hat{a}_{t+1} = \rho_A\hat{a}_t$. Note that this system is referred to as Equation 6 in the previous subsection.

Our solution reads as:

$$\hat{c}_t = F_c\hat{k}_{t-1} + G_c\hat{a}_t, \tag{20}$$

$$\hat{k}_t = F_k\hat{k}_{t-1} + G_k\hat{a}_t. \tag{21}$$

where F_c, F_k, G_c, G_k are unknown coefficients to be determined. To obtain these unknowns, we must inject these two equations in the previous system. Let us start with the first equation:

$$\begin{aligned} -\hat{k}_t + k_1\hat{a}_t + k_2\hat{k}_{t-1} - k_3\hat{c}_t &= 0, \\ -\left(F_k\hat{k}_{t-1} + G_k\hat{a}_t\right) + k_1\hat{a}_t + k_2\hat{k}_{t-1} - k_3\left(F_c\hat{k}_{t-1} + G_c\hat{a}_t\right) &= 0, \\ \underbrace{[k_2 - k_3F_c - F_k]}_{\text{condition (1)}}\hat{k}_{t-1} + \underbrace{[k_1 - k_3G_c - G_k]}_{\text{condition (2)}}\hat{a}_t &= 0. \end{aligned}$$

$$\begin{aligned} -F_k\hat{k}_{t-1} - G_k\hat{a}_t + k_1\hat{a}_t + k_2\hat{k}_{t-1} - k_3F_c\hat{k}_{t-1} - k_3G_c\hat{a}_t &= 0, \\ \left(-F_k + k_2 - k_3F_c\right)\hat{k}_{t-1} + \left(k_1 - k_3G_c - G_k\right)\hat{a}_t &= 0, \end{aligned}$$

Note that we must keep injecting these equations until we obtain a system made of $\hat{k}_{t-1}$ and $\hat{a}_t$. Following those steps, we obtain two conditions on parameters. For the second equation, we do exactly the same as

the previous equation to obtain a system with variables $(\hat{k}_{t-1}, \hat{a}_t)$:

$$\begin{aligned}
-\hat{c}_t + E_t\hat{c}_{t+1} - c_1 E_t\hat{a}_{t+1} - c_2\hat{k}_t &= 0,\\
-\hat{c}_t + F_c\hat{k}_t + G_c E_t\hat{a}_{t+1} - c_1 E_t\hat{a}_{t+1} - c_2\hat{k}_t &= 0,\\
-\hat{c}_t + (F_c - c_2)\,\hat{k}_t + \rho_A\,(G_c - c_1)\,\hat{a}_t &= 0,\\
-F_c\hat{k}_{t-1} - G_c\hat{a}_t + (F_c - c_2)\left(F_k\hat{k}_{t-1} + G_k\hat{a}_t\right) + \rho_A\,(G_c - c_1)\,\hat{a}_t &= 0,\\
\underbrace{[-F_c + (F_c - c_2)\,F_k]}_{\text{condition (3)}}\hat{k}_{t-1} + \underbrace{[(F_c - c_2)\,G_k + \rho_A\,(G_c - c_1) - G_c]}_{\text{condition (4)}}\hat{a}_t &= 0.
\end{aligned}$$

If we find the values of F_c, G_c, F_k, G_k that satisfy conditions (1-4), then we have found the solution of our model. Consider these four conditions:

$$\begin{aligned}
&\text{condition (1)} : 0 = k_2 - k_3 F_c - F_k\\
&\text{condition (2)} : 0 = k_1 - k_3 G_c - G_k\\
&\text{condition (3)} : 0 = -F_c + (F_c - c_2)\,F_k\\
&\text{condition (4)} : 0 = -G_c + (F_c - c_2)\,G_k + \rho_A\,(G_c - c_1)
\end{aligned}$$

Combining condition (1) and (3), we obtain the following problem:

$$-k_3 F_c^2 + (k_3 c_2 - 1 + k_2)\,F_c - k_2 c_2 = 0.$$

Note that this problem has two roots, the usual practice in macroeconomics is to pick the stable root of the system, i.e., $|F_k| = |k_2 - k_3 F_c| < 1$. Based on the stable root, we can get the remaining unknowns recursively as follows:

$$\begin{aligned}
F_k &= k_2 - k_3 F_c\\
G_c &= \frac{(F_c - c_2)\,k_1 - \rho_A c_1}{(F_c - c_2)\,k_3 + (1 - \rho_A)}\\
G_k &= k_1 - k_3 G_c
\end{aligned}$$

To get a presentation of the solution close to the one of Dynare, the problem can be recast as follows:

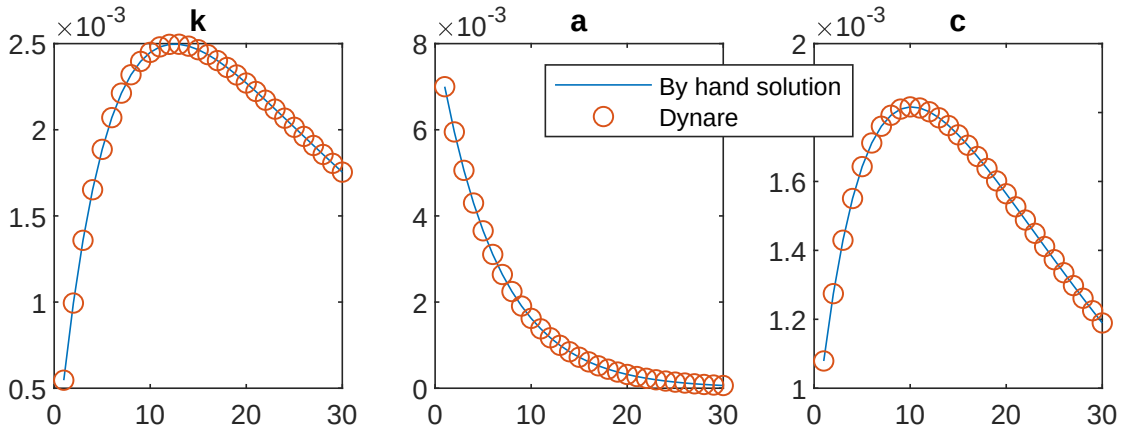
$$\begin{bmatrix} \hat{k}_t \\ \hat{a}_t \\ \hat{c}_t \end{bmatrix} = \begin{bmatrix} F_k & G_k\rho_A & 0 \\ 0 & \rho_A & 0 \\ F_c & G_c\rho_A & 0 \end{bmatrix} \begin{bmatrix} \hat{k}_{t-1} \\ \hat{a}_{t-1} \\ \hat{c}_{t-1} \end{bmatrix} + \begin{bmatrix} G_k \\ 1 \\ G_c \end{bmatrix} \epsilon_t. \tag{22}$$

Which we will write as in the next section:

$$x_t = F x_{t-1} + G\epsilon_t, \tag{23}$$

where $x_t = [\hat{k}_t, \hat{a}_t, \hat{c}_t]'$.

MATLAB box 1 compareRBC.m



3. Dynare Programming

We want to write the model into a form Dynare can interpret. In the first part of the Dynare code, we need to define the endogenous variables which appear in the model. The linear model has three equations and three endogenous variables: $\hat{c}_t$, $\hat{k}_t$ and $\hat{a}_t$. Code names omit hats, but still represent log deviations rather than levels. The syntax for writing this in Dynare is var variable1 variable2 ...; with the line ending in a semicolon. For our model, we write:

```
var k a c;
```

Next we write the parameters of the model. In our model, there are four structural parameters (α, β, δ and ρ_A) plus two steady state variables present in the linear model ($\bar{C}$ and $\bar{K}$), giving six parameters to declare. The syntax for writing this in Dynare is parameters parameter1 parameter2, ... ; with the line ended by a semicolon. Here we write:

```
parameters  alpha beta delta rho_A Css Kss;
```

Next, declare the innovation η_t^A. The persistent productivity process itself is an endogenous variable in Dynare. The syntax for writing this in Dynare is varexo variable1 variable2, ... ; with the line ending in a semicolon. For our set-up, we write:

```
varexo e_a;
```

Finally, we give the parameters values using our calibration in the parameter block below. The syntax for assigning parameter values is parameter = value; which is similar to normal MATLAB code. We write:

```
alpha         = 0.40;
beta             = 0.99;
delta             = 0.025;
rho_A         = 0.95;
Kss         = ((1/beta-(1-delta))/alpha)^(1/(alpha-1));
Css         = Kss^alpha - delta*Kss;
```

We now write the equations of the linearised model. Those equations are usually written in the same way as they appear in the paper. The equation section starts with model (linear); and ends with end;.[3] A few syntax considerations:

1. Variables which are multiplied (divided) by a parameter are separated from the parameter by the symbol * (/).
2. Contemporaneous variables do not have a time subscript in the Dynare code. Lagged variables x_{t-i} are written as x(-i), while expectational variables $\mathbb{E}_t x_{t+i}$ are written as x(+i).
3. Equations end with a semicolon.

The equations in Dynare syntax then are:

```
model(linear);
        [name='Budget constraint']
        k = Kss^(alpha-1)*a + (Kss^(alpha-1)*alpha + (1-delta))*k(-1) - Css/Kss*c;
        [name='Euler equation']
        c = c(+1) - beta*alpha*Kss^(alpha-1)*(a(+1) + (alpha-1)*k);
        [name='Productivity shock']
        a = rho_A*a(-1)+e_a;
end;
```

After specifying the equations, we define the shock, and set its standard deviation. We begin the section with a tag shocks; and end it with end;. The syntax for specifying each shock in Dynare is var shock = shock variance;. For the model, the section is:

```
shocks;
        var e_a;  stderr .007;
end;
```

Compute the steady state, inspect equation residuals, then check local determinacy with the Blanchard–Kahn conditions. In deviations, the steady state is zero. Use steady, resid and check before first-order simulation. In Dynare, we add:

[3] Note that the tag model (linear); includes the option 'linear' because the model is already linearized. For a nonlinear model, just write model ;.

```
check;
```

Finally, we solve the model and perform stochastic simulations. To do so, we employ the function stoch_simul(options=value) variable 1 variable 2;. Here we want a 30-period system response of consumption, capital and productivity after a technological shock. We write accordingly:

```
stoch_simul(order=1,irf=30) c k a;
```

There are many options available for the stoch_simul() function, see directly the Dynare user guide Adjemian et al. [2011] for details about options.

3.1 Simulating the economy

To run the model, the code has to be saved in a .mod file (mine is called basicRBC.mod). Set MATLAB's current directory to session2/codes, where the model file is located, and add your installed Dynare MATLAB directory to the path. To initiate Dynare, in MATLAB's command line, type:

console

```
dynare basicRBC
```

It is possible to insert more arguments in the Dynare command (see the manual for details). After pressing enter, you will then see Dynare at work. What it is doing is reading in the .mod contents, recognising the model's details, and solving for the linear rational expectations solution. A key advantage of Dynare compared to other alternatives is that Dynare creates and fills the matrices with the appropriate parameters to solve the linear rational expectations model and prints and saves all results. A large amount of information appears in the MATLAB command window, including the stability of the system and the model's second-order moment statistics.

console response

```
Using 64-bit preprocessor
Starting Dynare (version 4.6.2, historical output).
Calling Dynare with arguments: none
Starting preprocessing of the model file ...
Found 3 equation(s).
Evaluating expressions...done
Computing static model derivatives (order 1).
Computing dynamic model derivatives (order 2).
Processing outputs ...
done
Preprocessing completed.
```

The console excerpts and plots in this section are historical illustrations, not a fresh Dynare 7 validation run. The numerical moments have been checked analytically against the displayed policy rule, which uses $\rho_A = 0.95$. The supplied simulation files use $\rho_A = 0.85$, so numerical results differ unless the calibration is aligned. The first console block reports preprocessing information.

Next, Dynare checks whether the Blanchard–Kahn condition is verified.[4]

console response

```
EIGENVALUES:
 Modulus Real Imaginary
0.95 0.95 0
0.9691 0.9691 0
1.042 1.042 0
1.262e+17 1.262e+17 0



There are 2 eigenvalue(s) larger than 1 in modulus
for 2 forward-looking variable(s)


The rank condition is verified.
```

Here, as the condition is verified, the simulation can continue.

Next, Dynare generates the variance-covariance matrix of the shocks. The elements on the diagonal correspond to the specified values of the shock's variance. Here, we only have one shock, so the matrix is 1x1:

console response

```
MATRIX OF COVARIANCE OF EXOGENOUS SHOCKS


Variables    e_a
e_a            0.000049
```

[4]This is a condition defining the stability condition of the model. As explained in Dynare website, the Blanchard and Kahn conditions state that in order to have a unique stable trajectory you need to have as many eigenvalues larger than 1 in modulus as you have forward looking variables (variables with leads in Dynare. If the lead is on two periods as in y(+2), it counts for 2 forward looking variables).

In addition, a matrix that pops up in the computation of the solution must have full rank.

These conditions can fail in two ways: with more or fewer unstable eigenvalues than forward-looking variables. This leads respectively to nonexistence of a stable trajectory or to an infinity of stable trajectories (indeterminacy). Dynare will only proceed with computation if there exists a unique stable trajectory. You can display the eigenvalues with the "check" command added to the line after "steady" (the command to compute the steady state of a model). If this condition does not hold, Dynare stops processing the code and prints an error message.

Next, we obtain the policy and transition functions. Recall that in a compact form, Dynare computes the solution of the log-linear model (called policy function) as follows:

$$\hat{x}_t = F\hat{x}_{t-1} + G\epsilon_t, \tag{24}$$

where $\hat{x}_t = [\hat{k}_t, \hat{a}_t, \hat{c}_t]$ is the vector of the endogenous variables which is a self-determined process based on previous realizations of $\hat{x}_{t-1}$ + the current realization of shocks $\epsilon_t = [\eta_t^A]$. In the printed table, the lagged-state rows provide the nonzero columns of F after transposition. The innovation row provides G after transposition.

console response

POLICY AND TRANSITION FUNCTIONS

	c	k	a
k(-1)	0.653600	0.969086	0
a(-1)	0.269310	0.066465	0.950000
e_a	0.283484	0.069963	1.000000

We can then rewrite our backward looking process as:

$$\begin{bmatrix} \hat{c}_t \\ \hat{k}_t \\ \hat{a}_t \end{bmatrix} = \begin{bmatrix} 0 & 0.653600 & 0.269310 \\ 0 & 0.969086 & 0.066465 \\ 0 & 0 & 0.950000 \end{bmatrix} \begin{bmatrix} \hat{c}_{t-1} \\ \hat{k}_{t-1} \\ \hat{a}_{t-1} \end{bmatrix} + \begin{bmatrix} 0.283484 \\ 0.069963 \\ 1 \end{bmatrix} [\eta_t]$$

Since consumption $\hat{c}_t$ is only forward looking in the Euler equation, its lag does not enter the decision rule. The first column of F is zero.

The next section of Dynare output reports the moments of the model, using the model solution and the specified standard errors of the exogenous shocks.

console response

THEORETICAL MOMENTS

VARIABLE	MEAN	STD. DEV	VARIANCE
c	0.0000	0.0248	0.0006
k	0.0000	0.0313	0.0010
a	0.0000	0.0224	0.0005

Dynare then reports a matrix of theoretical correlations across the endogenous variables. These values are constructed from the variance-covariance matrix of the endogenous variables, which Dynare does not output to the command window.

console response

MATRIX OF CORRELATIONS

Variables	c	k	a
c	1.0000	0.9862	0.7516
k	0.9862	1.0000	0.6319
a	0.7516	0.6319	1.0000

Next, Dynare shows coefficients of autocorrelation up to the fifth order, that is, the correlation of an endogenous value with p-order lags of itself, $corr(\hat{z}_t, \hat{z}_{t-p})$ where $p = 1, 2, 3, 4, 5$.

console response

COEFFICIENTS OF AUTOCORRELATION

Order	1	2	3	4	5
c	0.9965	0.9917	0.9858	0.9788	0.9709
k	0.9992	0.9969	0.9933	0.9884	0.9823
a	0.9500	0.9025	0.8574	0.8145	0.7738

Under this historical calibration, capital is more persistent than consumption. Consumption is smoother than capital, but not constant. These are model implications, not empirical validation. Match definitions, units and filtering before comparing model moments with data.

Finally, Dynare also plots the system impulse response functions (IRF), *i.e.* the response of endogenous variables after the positive realization of each exogenous shock. Since our model only includes one shock, Figure 1 draws the response of consumption $\hat{c}_t$, physical capital $\hat{k}_t$ and productivity $\hat{a}_t$ after an innovation of 0.007 in log units, approximately 0.7%. The stored plot is historical. Its calibration should not be inferred solely from the adjacent console excerpt.

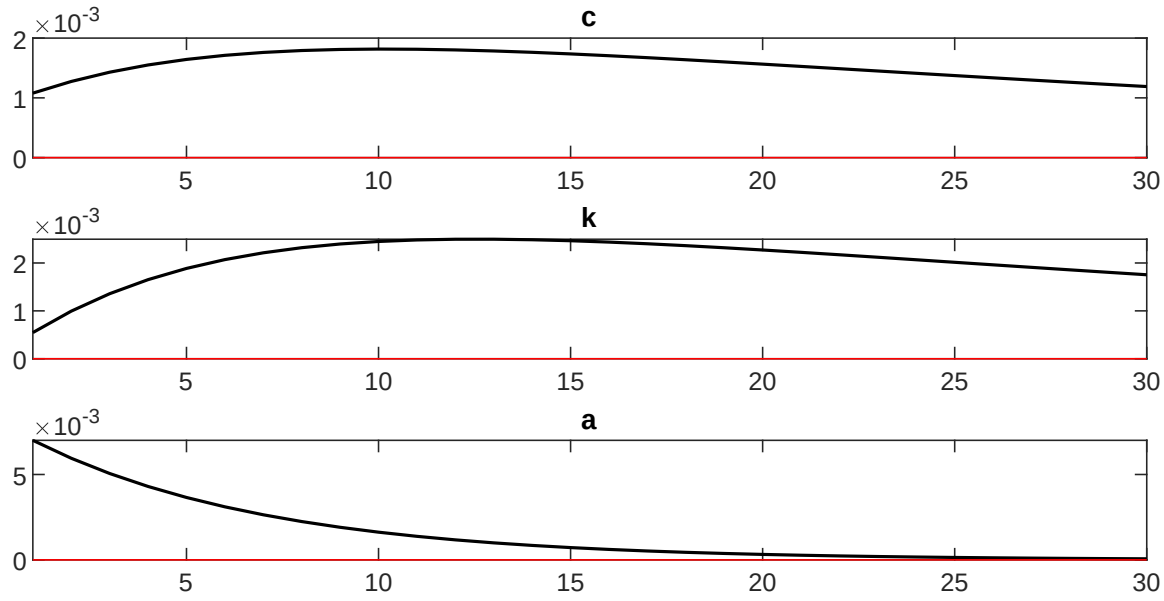


Figure 1: Historical impulse responses to a productivity innovation.

Stored in	Description
*oo_*	Structure of the output of stoch_simul function
oo_.exo_simul	Simulated series of the exogenous variables (if simulating).
oo_.endo_simul	Simulated series of the endogenous variables (if simulating).
oo_.dr	Stored information about variables and policy function.
oo_.dr.order_var	Declaration-order indices listed in decision-rule order.
oo_.dr.inv_order_var	Decision-rule positions listed in declaration order.
oo_.dr.ghx	State coefficient matrix in decision-rule order.
oo_.dr.ghu	Innovation coefficient matrix in decision-rule order.
oo_.steady_state	Values of the endogenous variables' steady-state.
oo_.variance_decomposition	The theoretical variance decomposition.
oo_.mean	The mean of endogenous variables.
oo_.var	The variance-covariance matrix.
oo_.autocorr	The autocorrelation matrices for endogenous variables, up to 5 lags.
oo_.irfs	Impulse responses of endogenous variables to exogenous shocks.
oo_.irfs.x_y	Impulse response of x to an innovation in y.
oo_.forecast	Forecasts of endogenous variables.
*M_*	Structure with information about the model.
M_.endo_names	The names of endogenous variables in declaration order.
M_.exo_names	The names of exogenous variables in declaration order.
M_.exo_nbr	The number of exogenous variables.
M_.endo_nbr	The number of endogenous variables.
options_	Structure with all the options which Dynare recognises.

Table 1: MATLAB's workspace after a stochastic simulation

Exercise 2: extending the simple model

Read basicRBC.mod and extend a working copy of the model.

1. The GDP, Y_t and investment I_t are not (yet) included in the model. Include them and draw their responses to a productivity shock. To do so, log-linearize their equations and find their steady state.

2. Check whether the investment-to-GDP ratio $\bar{I}/\bar{Y}$ and $\bar{C}/\bar{Y}$ are consistent with the data (use 20% for investment and 60% for consumption as rough motivating OECD benchmarks, not new estimates for a specified sample). Why is there such a huge gap? How could you improve the RBC model?

3. Change the calibration to obtain a non-stationary system response. What is Dynare telling you in the command window? Check the steady state, eigenvalues and rank condition separately. Does increasing α above one preserve a valid optimum, not merely a formal solution to the first-order conditions?

Exercise 3: extension: habits in consumption

Standard consumption-based models have difficulty accounting for the observed equity premium with conventional preference parameters. In asset pricing theory, habit formation became one candidate mechanism for changing how consumption risk is priced. Contributions include Constantinides [1990] and Abel [1990]. Aggregate consumption is typically smoother over the business cycle than investment and output. Habit formation offers one possible mechanism for consumption inertia: higher consumption raises a future reference level. Smooth aggregate consumption alone does not identify habits, however. Distinguishing this mechanism from other sources of persistence requires additional evidence.
Keep logarithmic utility, $\mathcal{U}(x) = \log x$, in both variants and require positive surplus consumption. Capital returns and feasibility are unchanged, but the marginal-utility equation changes.

1. First consider *external* habits. Utility depends on current consumption relative to a lagged aggregate reference:

$$\mathcal{U}\left(C_{jt} - hC_{t-1}\right) \tag{25}$$

   where $h \in [0, 1)$ measures habit strength. The reference is aggregate consumption C_{t-1}, without an individual subscript j. Each household takes this reference as given when choosing consumption. This mechanism is called *catching up with the Joneses*, because utility depends on consumption relative to a social reference.[a]

   (a) The marginal value of consumption is $\lambda_t = \mathcal{U}'\left(C_{jt} - hC_{t-1}\right)$.

   (b) Find the steady state.

   (c) Linearize the expression. It might be useful to split the problem with $\lambda_t = \mathcal{U}'\left(x_t\right)$ and $x_t = C_t - hC_{t-1}$ to apply the chain rule around the steady state.

   (d) How does persistence change relative to $h = 0$ at the same calibration? Re-solve the model rather than assume a universal ranking.

2. Let us now assume internal habits, denoted $\mathcal{U}\left(C_{jt} - hC_{jt-1}\right)$.

   (a) The marginal value of consumption is $\lambda_t = \mathcal{U}'\left(C_{jt} - hC_{jt-1}\right) - h\beta E_t\left[\mathcal{U}'(C_{jt+1} - hC_{jt})\right]$.

   (b) Find the steady state.

   (c) Linearize the expression. It might be useful to split again the problem with $\lambda_t = \mathcal{U}'\left(x_t\right) - \beta h E_t\left[\mathcal{U}'(x_{t+1})\right]$ and $x_t = C_t - hC_{t-1}$.

   (d) How does persistence change relative to $h = 0$ at the same calibration? Re-solve the model rather than assume a universal ranking.

[a] See Abel [1990] for the asset-pricing mechanism.

Exercise 4: Demand shocks

Assume now that our economy has a government. The government finances public spending by charging a lump-sum tax T_{jt} on households. The total amount of nominal public spending is $P_t G_t$. The government budget constraint is:

$$P_t G_t = \int_0^1 T_{jt} \mathrm{d}j \quad (=T_t). \tag{26}$$

Since households are identical in the symmetric equilibrium, we can write T_t directly. The budget constraint of the representative household is now:

$$C_{jt} + I_{jt} = A_t K_{jt-1}^{\alpha} - \frac{T_{jt}}{P_t}, \tag{27}$$

The aggregate resource constraint, or expenditure identity, becomes:

$$Y_t = C_t + I_t + G_t. \tag{28}$$

Government spending is exogenous to private decisions. In the absence of an explicit model of political cycles, spending varies around a fixed fraction of steady-state output:

$$G_t = \bar{G}\varepsilon_t^G = g^y \bar{Y} \varepsilon_t^G, \tag{29}$$

where spending follows an $AR(1)$ process:

$$\log\left(\varepsilon_t^G\right) = \rho_G \log\left(\varepsilon_{t-1}^G\right) + \eta_t^G, \tag{30}$$

where $0 \leq \rho_G < 1$ is the autocorrelation coefficient of the demand shock and η_t^G is an innovation distributed normally with zero mean and finite variance σ_G^2 such that $\eta_t^G \sim \mathcal{N}(0, \sigma_G^2)$. Parameter g^y is the spending-to-GDP ratio. We borrow the values of Smets and Wouters [2007] and assume that $\rho_G = 0.97$, $\sigma_G = 0.053$ and $g^y = 0.20$. Implicitly, in steady state, the shock is normalized to one $\bar{\varepsilon}^G = 1$.

1. Compute the new steady state of the model. *Tips:* (i) define a new parameter, $g^y = 0.2$, which affects the new steady state. (ii) since the budget is balanced, you can substitute real taxes T_t/P_t by spending G_t.

2. Calibrate the spending process and plot the impulse responses to a spending innovation.

3. Compare the response with a demand-led Keynesian mechanism. What can output do on impact when labour and installed capital are fixed? Discuss forward-looking responses to lump-sum taxes. Distinguish this wealth effect from Ricardian equivalence, which concerns tax versus debt timing and is not tested by this balanced-budget experiment. Which responses would you compare with data?

4. Assume independent technology and spending innovations. Analyse the variance decomposition. How does the main driver of output depend on shock variances and persistence?

5. Is this RBC model now able to replicate another business cycle fact which says output is more volatile than consumption?

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